NEW ERA UNIVERSITY

College of Computer Studies / Information Technology

NEU LIBRABASE
VISITOR LOG SYSTEM

System Documentation — Final Version

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1. Project Overview & Purpose

1.1 Introduction

The NEU LibraBase Visitor Log System is a full-stack web application developed for New Era University's library to digitize and streamline the visitor management process. The system replaces traditional paper-based logbooks with a modern, real-time digital platform that records visitor check-ins, tracks library usage patterns, and provides administrators with analytical insights.

This system was developed as an academic project under the Information Management course, demonstrating the practical application of web technologies, database management, Google OAuth 2.0 authentication, role-based access control, and cloud deployment via Vercel and GitHub.

1.2 Project Objectives

- Provide a digital platform for recording library visitor check-ins and check-outs with time tracking.
- Enable secure user registration and authentication using institutional email addresses (@neu.edu.ph).
- Implement Google OAuth 2.0 for seamless one-click institutional login.
- Support role-based access control with Visitor, Admin, and dual-role accounts.
- Allow visitors to update their student number, user type, and college directly on the check-in form.
- Display a detailed check-in summary card after every successful visit log.
- Give library administrators a real-time dashboard with analytics and visitor management tools.
- Deploy the system on a live cloud platform accessible to all NEU students and staff.

1.3 Target Users

- Visitors — NEU students, faculty, staff, and professors who visit the library and use the system to log their visits.
- Administrators — Library staff and designated personnel who manage and monitor visitor records and analytics.
- Dual-Role Users — Accounts like christian.adlawan@neu.edu.ph and jcesperanza@neu.edu.ph that have both visitor and admin access.

1.4 Live Deployment

Live URL: https://neu-library-visitor-log-ochre.vercel.app

GitHub Repository: https://github.com/Chandara4you/neu-library-visitor-log

2. System Features

2.1 Authentication & Access Control

Login Methods

- Email and Password Login — using institutional @neu.edu.ph email addresses via the Email tab.
- Student Number Login — visitors can log in using their student number and password via the Student No. tab.
- Google OAuth Login — one-click Sign in with Google using NEU institutional accounts. Restricted to @neu.edu.ph email addresses only. Any non-NEU account is automatically rejected.

Role-Based Access Control

- Regular Visitor Role — access to the Visitor Portal (Log a Visit only).
- Admin Role — access to the full Admin Panel with analytics, visitor management, and visit logs.
- Dual-Role — accounts registered in both the users and admins tables are shown a Role Chooser screen after Google login, allowing them to pick which portal to enter.
- Change Role Button — both the Visitor Portal and Admin Panel have a gold-colored Change Role button in the sidebar footer (beside Sign Out). This lets dual-role users switch portals without logging out. The button only appears for accounts that have both roles.
- Account Blocking — administrators can block visitor accounts. Blocked accounts cannot log in.

Session & Security

- Inactivity Timeout — sessions auto-expire after 15 minutes of inactivity. Time-out is recorded in the database before redirecting to login.
- Manual Sign Out — time-out is recorded automatically on sign out.
- Domain restriction — enforced in auth-callback.html after every Google login.
- Auto-provisioning — new Google login users are automatically given a visitor account on first sign-in.

2.2 Visitor Portal

Log a Visit Form

- Pre-filled fields: Full Name and Email from the visitor's profile (read-only).
- Editable Student Number — visitors can type or update their student number each visit. If updated, it is automatically saved back to their profile in Supabase for future visits.
- Editable User Type — dropdown: Student, Faculty, Staff, Professor. Changes are saved to profile.
- Editable College/Department — dropdown with all 12 NEU colleges. Changes are saved to profile.
- 10 Purpose of Visit tiles: Reading, Studying, Research, Computer, Group Study, Borrowing Books, Returning Books, Internet Access, Consultation with Librarian, Others.

- Two-column layout: form on the left, Library Hours and Library Reminders panel on the right.

Check-In Summary Card

- After clicking Check In to Library, a modal overlay appears with a full summary card.
- Summary card displays: visitor avatar initials, full name, email, student number, user type, college, purpose of visit, and time in.
- A Continue to Portal button closes the modal and resets the form, ready for the next check-in.

2.3 Admin Panel

Dashboard

- 5 real-time stat cards: Visitors Today, This Week, This Month, Total Visitors, Top Purpose.
- 4 analytics charts: Visitors Per Day (last 14 days bar chart), Purpose Distribution (doughnut), Visitors by College (bar), Peak Hours Heatmap (day × hour grid).

Visitor Management

- Full table of all registered visitors with search by name, email, student number, or college.
- Filter by user type and college/department.
- Block and Unblock visitors — blocked visitors cannot log in.

Visit Logs

- Complete visit records with Time In and Time Out columns.
- Active visits show 'Active'; completed visits show the exact check-out time.
- Search, filter by date range and purpose, bulk delete, and CSV export.
- Pagination for large datasets.

2.4 Registration

- Two-step registration: Step 1 — Personal Info (name, student number, user type, college). Step 2 — Account Credentials (institutional email, password, confirm password).
- @neu.edu.ph email validation enforced. Duplicate email and student number detection.
- Note: Google login users are auto-provisioned and do not need to register manually.

2.5 UI/UX

- System name: NEU LibraBase Visitor Log System across all pages.
- Light and Dark mode toggle on all 5 pages, persisted via localStorage.
- NEU branding: crimson red (#8B0000) and gold (#D4AF37) color scheme throughout.
- Glassmorphism sidebar, smooth animations, toast notification system.
- Responsive design for desktop and mobile.

3. Technologies Used

Category	Technology	Purpose
Frontend	HTML5, CSS3, JavaScript (ES6+)	Core structure, styling, interactivity
Typography	Google Fonts — Playfair Display, DM Sans	System-wide typography
Charts	Chart.js v4.4.0	Admin dashboard analytics
Database	Supabase (PostgreSQL)	Database, auth, real-time data, RLS
Authentication	Supabase Auth + Google OAuth 2.0	Email/password and Google login
Google OAuth	Google Cloud Console — OAuth 2.0 Client	Sign in with Google functionality
Deployment	Vercel	Static hosting with GitHub auto-deploy
Version Control	Git + GitHub	Source code and deployment pipeline
IDE	Visual Studio Code + Live Server	Local development
Architecture	Static HTML/CSS/JS — no build tools	Simple, serverless frontend

4. Database Schema

The system uses PostgreSQL hosted on Supabase with Row Level Security (RLS) enabled on all tables.

4.1 users — Visitor Accounts

Column	Type	Description
id	UUID (PK)	Auto-generated unique identifier
full_name	TEXT	Visitor's full name
email	TEXT UNIQUE	Institutional email (@neu.edu.ph)
student_number	TEXT UNIQUE	Student number (editable per visit, optional)
user_type	TEXT	Student / Faculty / Staff / Professor
college	TEXT	College or department
password_hash	TEXT	Base64 password or 'google-oauth' for OAuth users
is_blocked	BOOLEAN	Whether account is blocked (default: false)
created_at	TIMESTAMPTZ	Account creation timestamp

4.2 visits — Visit Records

Column	Type	Description
id	UUID (PK)	Auto-generated unique identifier
user_id	UUID (FK)	References users.id — cascade delete
full_name	TEXT	Name at time of visit
email	TEXT	Email at time of visit
student_number	TEXT	Student number at time of visit
user_type	TEXT	User type at time of visit
college	TEXT	College at time of visit
purpose	TEXT NOT NULL	Purpose of visit (required)
time_in	TIMESTAMPTZ	Check-in timestamp
time_out	TIMESTAMPTZ	Check-out timestamp (null = still active)
date	DATE	Visit date

4.3 admins — Administrator Accounts

Column	Type	Description

id	UUID (PK)	Auto-generated unique identifier
name	TEXT	Administrator's full name
email	TEXT UNIQUE	Admin email address
password_hash	TEXT	Password or 'google-oauth' for OAuth admins
created_at	TIMESTAMPTZ	Account creation timestamp

4.4 Pre-seeded Admin Accounts

Name	Email	Login Method	Role
Library Admin	admin@neu.edu.ph	Email / Password (Admin@NEU2024)	Admin only
Prof. JC Esperanza	jcesperanza@neu.edu.ph	Google OAuth	Admin + Visitor (dual-role)
Christian Adlawan	christian.adlawan@neu.edu.ph	Google OAuth	Admin + Visitor (dual-role)

4.5 Colleges — Reference List

The system includes 12 colleges available in all dropdown selectors:

- College of Informatics and Computing Studies
- College of Informatics and Computer Studies
- College of Arts and Sciences
- College of Business Administration
- College of Computer Studies
- College of Criminal Justice Education
- College of Education
- College of Engineering
- College of Nursing
- College of Tourism and Hospitality Management
- Graduate School
- Senior High School

5. User Guide

5.1 Accessing the System

Open a browser and go to the live URL:

```
https://neu-library-visitor-log-ochre.vercel.app
```

5.2 Registering an Account (Manual)

1. Click Get Started from the landing page.
2. Step 1: Enter Full Name, Student Number (optional), User Type, and College.
3. Step 2: Enter your @neu.edu.ph email, password (min. 8 characters), and confirm it.
4. Click Create Account. You will be redirected to the login page.

Note: Google login users are auto-provisioned and do not need to register manually.

5.3 Logging In

Email and Password

5. Click the Email tab on the login page.
6. Enter your @neu.edu.ph email and password. Click Sign In.
7. Admin accounts go to the Admin Panel; visitor accounts go to the Visitor Portal.

Student Number

8. Click the Student No. tab.
9. Enter your student number and password. Click Sign In.

Google Sign In

10. Click Sign in with Google.
11. Select your @neu.edu.ph Google account from the picker.
12. If you have both visitor and admin roles, the Role Chooser screen appears — select your desired portal.
13. Non-@neu.edu.ph accounts are rejected automatically.

Note: First-time Google login users are automatically given a visitor account — no manual registration needed.

5.4 Logging a Library Visit

14. After login, you are on the Visitor Portal showing the Log a Visit form.
15. Your Full Name and Email are pre-filled and read-only.
16. Student Number — type or update your student number. If you update it, it is saved to your profile automatically.
17. Select your User Type from the dropdown (saved to profile if changed).
18. Select your College/Department (saved to profile if changed).
19. Select your Purpose of Visit from the 10-tile grid.
20. Click Check In to Library.

21. A Check-In Summary Card appears showing your name, email, student number, user type, college, purpose, and time in.
22. Click Continue to Portal to close the summary and return to the form.

5.5 Switching Roles (Dual-Role Users Only)

23. In the sidebar footer, click the gold Change Role button (beside Sign Out).
24. A role chooser overlay appears — select Visitor Portal or Admin Panel.
25. You are redirected to the chosen portal immediately.

Note: The Change Role button is only visible to accounts that have both visitor and admin roles.

5.6 Signing Out

Click Sign Out in the sidebar footer. Your time-out is recorded automatically before the session ends. If you are inactive for 15 minutes, the system times out automatically and records the time-out.

6. Admin Guide

6.1 Accessing the Admin Panel

Log in with an admin account. You will be redirected to admin.html automatically. The Admin Panel has three sections: Dashboard, Visitors, and Visit Logs.

6.2 Dashboard

- 5 stat cards — Visitors Today, This Week, This Month, Total Visitors, Top Purpose.
- Visitors Per Day chart — last 14 days bar chart.
- Purpose Distribution chart — all-time doughnut breakdown.
- Visitors by College chart — top colleges bar chart.
- Peak Hours Heatmap — visit frequency by day of week and hour.

6.3 Visitor Management

26. Click Visitors in the sidebar.
27. Search by name, email, student number, or college.
28. Filter by user type and college.
29. Click Block to prevent a visitor from logging in. Click Unblock to restore access.

6.4 Visit Logs

30. Click Visit Logs in the sidebar.
31. Records show: #, Date, Name, Student No., College, Purpose, Time In, Time Out.
32. Time Out shows the check-out time or 'Active' for ongoing visits.
33. Use search, date range, and purpose filters to narrow results.
34. Click the trash icon to delete individual records.
35. Check multiple rows then click Delete Selected for bulk delete.
36. Click Export CSV to download all filtered records.

6.5 Switching to Visitor Portal

Admin users with dual roles can click Change Role in the sidebar footer to switch to the Visitor Portal without logging out.

7. Google OAuth Setup Guide

7.1 Google Cloud Console

37. Go to console.cloud.google.com → create project NEU LIBRARY VISITOR LOG.
38. Go to Google Auth Platform → Get Started → complete consent screen setup.
39. Go to Clients → Create Client → Web application → name: NEU Library Log Web.
40. Add Authorized JavaScript Origins:

```
https://neu-library-visitor-log-ochre.vercel.app
http://127.0.0.1:5500
```

41. Add Authorized Redirect URIs:

```
https://neu-library-visitor-log-ochre.vercel.app/auth-callback.html
https://xoznxbqfgbkynmjhwqrq.supabase.co/auth/v1/callback
http://127.0.0.1:5500/neu-library-system/auth-callback.html
```

42. Save and copy Client ID and Client Secret.
43. Go to Audience → click Publish App → Confirm (removes test-user restriction so all @neu.edu.ph users can sign in).

7.2 Supabase Configuration

44. Go to Supabase Dashboard → Authentication → Sign In / Providers → Google → Enable.
45. Paste Client ID and Client Secret. Click Save.
46. Go to Authentication → URL Configuration.
47. Set Site URL to:

```
https://neu-library-visitor-log-ochre.vercel.app
```

48. Add Redirect URLs:

```
https://neu-library-visitor-log-ochre.vercel.app/auth-callback.html
http://127.0.0.1:5500/neu-library-system/auth-callback.html
```

49. Click Save.

7.3 OAuth Flow

50. User clicks Sign in with Google → Supabase signInWithOAuth() is called.
51. User selects @neu.edu.ph account from Google picker.
52. Google redirects to auth-callback.html with access token.
53. auth-callback.html checks domain — non-@neu.edu.ph accounts are rejected.
54. System checks admins and users tables simultaneously.
55. Dual-role → Role Chooser shown. Admin only → Admin Panel. Visitor only → Visitor Portal.
56. First-time users are auto-provisioned a visitor account.

8. Deployment Guide

8.1 Initial Deployment

57. Create GitHub repository: neu-library-visitor-log (public, with README).
58. Open project in VS Code terminal and run:

```
git init
git add .
git commit -m "Initial commit"
git branch -M main
git remote add origin https://github.com/Chandara4you/neu-library-visitor-log.git
git pull origin main --allow-unrelated-histories
git push -u origin main
```

59. Go to vercel.com → Add New Project → Import neu-library-visitor-log → Framework: Other → Deploy.
60. Live URL is assigned automatically.

8.2 Updating the Deployed App

Every time you make changes locally, run these three commands. Vercel auto-redeploys within 30–60 seconds:

```
git add .
git commit -m "describe your change"
git push
```

If push is rejected (remote has newer commits), run this first then push again:

```
git pull origin main --allow-unrelated-histories
```

8.3 vercel.json

```
{ "cleanUrls": false }
```

This prevents Vercel from stripping .html extensions from URLs, which is required since all pages use explicit .html file links.

9. File Structure & Developer Notes

9.1 File Structure

```
neu-library-system/
├── index.html              ← Landing page
├── login.html              ← Login (Email, Student No., Google OAuth)
├── register.html           ← Two-step visitor registration
├── visitor.html            ← Visitor portal (Log a Visit + summary card)
├── admin.html              ← Admin panel (Dashboard, Visitors, Logs)
├── auth-callback.html      ← Google OAuth handler & role router
├── schema.sql              ← Full Supabase database schema
├── vercel.json             ← Deployment config
├── New Era Library Logo.png ← Transparent PNG logo
├── css/style.css           ← Global stylesheet
├── js/auth.js              ← Auth helpers
├── js/dashboard.js         ← Admin charts
├── js/supabase.js          ← Supabase client
└── assets/logo.svg         ← SVG logo asset
```

9.2 Key Implementation Details

Editable Student Number with Profile Sync

The student number field on the Log a Visit form is now editable. When a visitor submits the form, the system compares the entered student number, user type, and college with their stored profile. If any value has changed, it updates the users table in Supabase and refreshes the localStorage session so changes persist on future visits.

Check-In Summary Card

After a successful visit insert, all form values (name, email, student number, type, college, purpose, time in) are captured and injected into the modal overlay. The summary card shows a generated avatar using the visitor's initials, a formatted grid of visit details, and a Continue to Portal button that resets the form.

Change Role Feature

Both the Visitor Portal and Admin Panel sidebars have a Change Role button in the footer, styled in gold to distinguish it from the Sign Out button. The button is hidden by default and only appears if a DB query confirms the current user also exists in the opposite role table. Clicking it shows an overlay with Visitor Portal and Admin Panel options.

Inactivity Timeout

A 15-minute timer runs on the Visitor Portal. Any user interaction (mouse, keyboard, scroll, touch) resets it. On timeout, the system records time_out for the active visit, clears localStorage, and shows a timeout overlay before redirecting to login.

Auto-provisioning for Google Users

When a @neu.edu.ph Google account logs in for the first time, auth-callback.html checks if a record exists in the users table. If not, it creates one automatically using the Google profile name and email. This means students can use the system immediately after clicking Sign in with Google without registering manually.

9.3 Known Limitations

- Passwords are stored as Base64 encoding — for demonstration purposes only. A production system should use bcrypt or Argon2.
- Supabase anon key is embedded in client-side HTML — acceptable with RLS properly configured.
- Google OAuth app must be in Production mode (not Testing) for all @neu.edu.ph users to log in without being manually added as test users.

— End of Documentation —

NEU LibraBase Visitor Log System | Christian Paulino Adlawan | New Era University 2025–2026